

World Health Organization: Advancing Healthcare-Associated Infection Surveillance

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I. Summary

On Monday, November 24th, I attended a webinar hosted by the World Health Organization (WHO), titled *From Data to Impact: Advancing Healthcare-Associated Infection Surveillance for Safer Care and a Healthier Future*. The webinar was held during WHO World Antimicrobial Resistance Awareness Week, presented by WHO directors from a variety of world regions, and covered topics like global strategy, HAI/AMR surveillance methods and elements, and systems integration. 10 directors of different branches spoke on HAI challenges, with some major speakers being directors of infection prevention, universal health coverage, infectious disease, epidemiology, and regional advisors from different regions including the Eastern Mediterranean, Europe, and North America.

Robust surveillance and data collection are critical to dealing with Healthcare-Associated Infection (HAI). Most HAIs are caused by antimicrobial-resistant (AMR) microorganisms and disproportionately affect low-income countries, placing financial strain on their health systems. There are about 45 different types of HAI, and most cases are completely preventable. Many HAIs are caused by antimicrobial-resistant microorganisms and so are often seen as closely linked. Common cases include urinary tract infection (UTI), bloodstream infection (BSI), and surgical site infection (SSI). The role of HAI/AMR surveillance is to aid in early detection, where system-integrated signs and symptoms can identify causative organisms of the infection. Surveillance helps develop antimicrobial susceptibility profiles for health systems, making improvements both nationally and worldwide when partnered with global organizations like the WHO. The Review on AMR (2014) hypothesizes that without proper surveillance and prevention strategy, AMR has the potential to kill more people than cancer by 2050. The WHO sees enhanced data systems as a strong tool to lead change in HAI and AMR. The purpose of this report is to address some of the challenges faced in combating AMR as a public health threat, discuss integration of surveillance systems, and review global insights available from the WHO.

II. Existing Research

The opening of the “antibiotic era” can generally be tied to the discovery of penicillin in 1928, according to Caselli and Purificato (2020). We now find ourselves in a new period, defined

as the “post-antibiotic era”, where more than 33,000 deaths from AMR bacterial infections, acquired during hospitalization, are reported each year in the European Union alone. Widespread and sometimes inappropriate use of antibiotics in humans, particularly in healthcare settings, has led to alarming levels of AMR. In the U.S., 1 in every 31 hospitalized patients acquire a HAI, according to a CDC study from 2018 (Magill et al., 2018). Gagneux-Brunon, et al. (2018) notes that HAIs are particularly difficult to predict and treat because they are both endogenous and exogenous in nature, meaning the infection originates from the patient’s unique flora but also from cross transmission in the healthcare setting where acquired.

Successful initiatives like the CDC’s 1995 *Get Smart* campaign focused on providing information to parents in particular, educating them on the appropriate and inappropriate use of antibiotics to treat common childhood infections, launching an awareness of the dangers of AMR. This method was highly influential, significantly reducing antibiotic prescriptions, and demonstrating a new sense of stewardship in healthcare workers and parents alike (Rosenberg, 2012). One study from 2018 reviews the potential role of vaccines for HAI prevention, expecting results of large, advanced phase trials for *C. difficile* and *S. aureus* vaccines available in the next five years (Gagneux-Brunon, et al., 2018). The effectiveness of a prevention method like this could decrease infections and their relative costs. While many strategies have been developed and implemented to combat AMR, our current sophisticated technology environment provides a strong foundation for vital data collection and informatics methods.

Comprehensive AMR surveillance is the driver of antibiotic stewardship guidelines. Tacconelli et al. (2018) highlight that in the absence of high-quality surveillance data, early detection of AMR infection trends and case groups is compromised and sets both hospital and public health responses back. Frost et al. (2021) note that “surveillance platforms at national and international levels are crucial for quantifying the burden of AMR...” and play a role in “policy and clinical decision making”. The article reviews existing AMR surveillance systems and discusses challenges presented when standardization is lacking, further limiting the possibility for effective responses. Current research shows us that successful national and international surveillance systems are dependent on standardizing data across settings.

III. Critical Analysis

I have coded data visualizations in various demos using R Studio to look at HAI prevalence across U.S. counties, so I approach the topic with enhanced interest in the WHO's global surveillance methods. Complex, large-scale datasets, whether hosting U.S. national data or another world region, require summary and refining of metrics for any meaningful analysis. This webinar effectively broke down how surveillance systems acquire that data, described how they are implemented, and explained the role of the kind of data analysis that I had done before.

Speakers shared WHO goals for both national-level and facility-level HAI/AMR surveillance, stating that as of 2023-2024, their data showed 50% of the world's healthcare facilities had some form of surveillance in place. With a complex issue like AMR, there must be a close relationship between global strategy and local implementation of surveillance systems. Global organizations like WHO are leaders of that global strategy and necessary for things like advising on harmonization of case definitions, ensuring consistent identification of diagnostic criteria, and guiding nations with data-driven recommendations on combating HAI. High-income countries like the EU and the U.S. track HAI and AMR surveillance simultaneously, because of the close link between the two. This stands as the most ideal local implementation, but it involves very sophisticated tracking systems which are generally not financially available to low-income countries. The WHO's guidance emphasized a new focus on granular, individual-level data collection, and assigning more importance to variables like infection origin, patient type, and even hospital unit type – whether or not recorded within a highly technical data system.

WHO advises all countries to establish both national-level and facility-level HAI and AMR surveillance systems, most importantly to provide early warning for detecting epidemic and pandemic-prone AMR pathogens, but also to better monitor antimicrobial utilization by patients. Section 5 in their *Global Action Plan and Monitoring Framework on Infection Prevention and Control (2024a)*, titled “Data for Action”, presents implementation of efficient data collection and analysis training as a new global strategy to keep all people accessing healthcare safe from HAI. The document presents actionable data items such as establishing hand hygiene compliance monitoring in healthcare facilities as a “key national indicator”, integrating HAI indicators in national health information, and ensuring “training and expertise in data collection, analysis, interpretation, and reporting”. *Surveillance of Health Care-Associated*

Infections at National and Facility Levels, a WHO practical handbook published in 2024, includes many resources and information for surveillance implementations. The handbook breaks down vital steps in the data cycle, from data collection to analysis, interpretation, and then finally communication. This resource highlights the importance of surveillance for AMR and HAI, helping nations and care facilities alike find meaning and improvement opportunities in collected data.

IV. The “One Health Perspective

What makes AMR a unique problem is its large-scale impact on all parts of life, examining from a One Health approach. McEwen and Collignon (2018) describe One Health as a framework that intersects health science professions to help people, animals, and our environment reach optimal health and stability. Though many drug-resistant microorganisms emerge out of clinical settings, the interconnected processes of human, animal, and environmental health all play a role in this complex issue.

In food-producing animals, antibiotics are often used to treat infections among herds and promote growth. In the case that a small group of animals develop an infection and become sick, for reasons of practicality and efficiency, the antibiotics will be administered to the entire herd preventing further infection and death, even if the majority of the group is not already infected. McEwen and Collignon (2018) argue this practice is inappropriately defined as "therapeutic" use by those in the food animal sector. Additionally, antibiotics are used for growth promotion in a long-term, low-dose medication somehow justified on economic grounds. Notably, the WHO advocates for the termination of administering antimicrobials for growth promotion. The practice of using antibiotics in food animals promotes the same selective pressures as there are in a hospital, resulting in resistant bacteria to arise. From there, these bacteria move back to humans in the form of our food, creating a human-animal cycle that is detrimental to all sectors. Antimicrobial residues are also inevitably constituents of human sewage, livestock manure and aquaculture, and treatments of such waste sources can only reduce some antimicrobials and microorganisms present. The remainder will then be introduced back into our soil and water, making the environment an additional critical pathway we need to monitor for AMR bacteria.

The need for sector-specific changes in human healthcare, animal food production, and environmental spaces is more urgent than ever. The goal is to unify and engage all stakeholders. If this is met, we are better off, so long as research, policy, and legal groups can guide the consistent change. In healthcare settings, this will require translating evidence into standards enforced among staff, aiming to practice better antimicrobial stewardship. “The WHO embraces a One Health approach to address antimicrobial resistance, and it calls on member countries to do the same when developing their own action plans.” (McEwen and Collignon, 2018).

V. Application & Reflection

Antimicrobial medicines are foundational to our modern medicine, so the emergence of pathogens resistant to treatment seriously threatens to undo decades of advancements in medical practice and public health. Most HAIs are increasingly caused by microorganisms that are already resistant to first-line drugs. These HAIs are often acquired during hospitalization and may ultimately present as “multi-resistant” (Masia & Dettori, 2022). The Lancet (2022) reported nearly 5 million AMR-related deaths globally in 2019. AMR is arguably the most urgent global public health issue we are experiencing, requiring attention from all world regions and joining the WHO in their prevention and infection control efforts. “AMR is a problem for all countries at all income levels. Its spread does not recognize country borders.” (WHO, 2023). This webinar communicated WHO next steps, addressing the need for global collaboration on integrated HAI/AMR reporting, requiring more technical discussions about where current efforts can be modified for the best outcomes.

According to the WHO, data is key for helping the world address gaps in diagnosis, treatment, and especially surveillance. Attending this informational webinar with WHO has given me new insight on global strategy for HAI and AMR, inspiring my own involvement in increasing attention on these issues. As an ambassador for HAI and AMR change, I hope to play a role in rewriting the narrative on antibiotic use. I can lead as my own advocate by educating myself and those close to me on proper infection prevention strategies and encouraging community-level action through engagement with local government and public health initiatives.

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